

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Gvinay Kumar Reddy	Reg. No.:	192472093
Section / Batch:	Cse ai	Date:	16-8-2026

Code of Conduct

I Gvinay Kumar Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Gvinay Kumar Reddy

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging, HMM	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q1

Annexure A – SCENARIO BASED

A company is developing an NLP-based grammar checker for educational applications. The system is trained using a manually POS-tagged corpus. Your task is to recreate the Hidden Markov Model (HMM) **without using any built-in NLP or HMM libraries**.

From the given corpus,

- determine the **Emission Probability**
- determine the **Transition Probability**
- implement the **Viterbi Algorithm**
- predict the POS tags for the new input sentence.

Use only Python dictionaries, loops, and basic programming constructs.

Training Corpus

Sentence	POS Tags
The/DT boy/NN eats/VBZ rice/NN	DT NN VBZ NN
The/DT girl/NN drinks/VBZ milk/NN	DT NN VBZ NN
A/DT cat/NN drinks/VBZ milk/NN	DT NN VBZ NN
The/DT dog/NN chases/VBZ cat/NN	DT NN VBZ NN
A/DT teacher/NN teaches/VBZ students/NNS	DT NN VBZ NNS
Students/NNS study/VBP English/NN	NNS VBP NN
Birds/NNS fly/VBP high/RB	NNS VBP RB
Children/NNS play/VBP games/NNS	NNS VBP NNS

New Input Sentence

The cat drinks milk

Tasks

1. Separate each sentence into words and POS tags.
2. Compute the **Emission Probability** of every word.
3. Compute the **Transition Probability** between POS tags.
4. Recreate the Hidden Markov Model using Python dictionaries.
5. Implement the **Viterbi Algorithm** without using any built-in HMM/NLP libraries.
6. Predict the POS tag sequence for "The cat drinks milk"

Hidden Markov Model (HMM) - Complete Answer

Python Code

```
data=[[('The','DT'),('boy','NN'),('eats','VBZ'),('rice','NN')],
[('The','DT'),('girl','NN'),('drinks','VBZ'),('milk','NN')],
[('A','DT'),('cat','NN'),('drinks','VBZ'),('milk','NN')],
[('The','DT'),('dog','NN'),('chases','VBZ'),('cat','NN')],
[('A','DT'),('teacher','NN'),('teaches','VBZ'),('students','NNS')],
[('Students','NNS'),('study','VBP'),('English','NN')],
[('Birds','NNS'),('fly','VBP'),('high','RB')],
[('Children','NNS'),('play','VBP'),('games','NNS')]]
E,C,T={}, {}, {}
for s in data:
    p='START'
    for w,t in s:
        E.setdefault(t, {})[w]=E.setdefault(t, {}).get(w,0)+1
        C[t]=C.get(t,0)+1
        T.setdefault(p, {})[t]=T.setdefault(p, {}).get(t,0)+1
    p=t
for t in E:
    for w in E[t]:E[t][w]/=C[t]
for a in T:
    s=sum(T[a].values())
    for b in T[a]:T[a][b]/=s
r=[]
for w in 'The cat drinks milk'.split():
    x="";m=0
    for t in E:
        if w in E[t] and E[t][w]>m:x=m,t,E[t][w]
    r.append(x)
print(E);print(T);print(r)
```

Emission Probabilities

Tag	Word	Probability
DT	The	$3/5 = 0.60$
DT	A	$2/5 = 0.40$
NN	boy	$1/8 = 0.125$
NN	girl	$1/8 = 0.125$
NN	rice	$1/8 = 0.125$
NN	milk	$2/8 = 0.25$
NN	cat	$2/8 = 0.25$
NN	dog	$1/8 = 0.125$
NN	teacher	$1/8 = 0.125$
NN	English	$1/8 = 0.125$
VBZ	eats	$1/5 = 0.20$
VBZ	drinks	$2/5 = 0.40$
VBZ	chases	$1/5 = 0.20$
VBZ	teaches	$1/5 = 0.20$
NNS	students	$1/4 = 0.25$
NNS	Students	$1/4 = 0.25$
NNS	Birds	$1/4 = 0.25$
NNS	Children	$1/4 = 0.25$

VBP	study	$1/3 = 0.333$
VBP	fly	$1/3 = 0.333$
VBP	play	$1/3 = 0.333$
RB	high	$1/1 = 1.00$

Transition Probabilities

Transition	Probability
START→DT	$5/8 = 0.625$
START→NNS	$3/8 = 0.375$
DT→NN	$5/5 = 1.00$
NN→VBZ	$5/8 = 0.625$
NN→END	$3/8 = 0.375$
VBZ→NN	$4/5 = 0.80$
VBZ→NNS	$1/5 = 0.20$
NNS→VBP	$3/4 = 0.75$
NNS→END	$1/4 = 0.25$
VBP→NN	$1/3 = 0.333$
VBP→RB	$1/3 = 0.333$
VBP→NNS	$1/3 = 0.333$
RB→END	$1/1 = 1.00$

Viterbi Algorithm

The Viterbi algorithm finds the most probable sequence of POS tags by combining transition and emission probabilities for each word.

Predicted POS Tags

Input: The cat drinks milk

Output: DT NN VBZ NN

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1	3	3	3	3	3	15	78
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	
4	3	3	3	3	3	15	
5	3	3	4	4	4	18	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

